

ERGONOMICS SAFETY STUDY GUIDE

Preventing Strain, Fatigue, and Musculoskeletal Injuries

Key Takeaway: Ergonomic injuries rarely happen suddenly. They develop slowly through daily habits. Small adjustments made early can prevent serious long-term injury.

Why Ergonomics Matters

Ergonomics is the practice of fitting the job to the worker rather than forcing the worker to adapt to poor posture, awkward movement, or inefficient workstation design. Poor ergonomics places unnecessary stress on the body and increases the risk of injury.

Common Ergonomic Injuries

- Back and neck strain caused by poor posture
- Shoulder, elbow, and wrist injuries from repetitive motion
- Tingling or numbness in hands or fingers
- Muscle fatigue, stiffness, and reduced range of motion

Posture and Neutral Positioning

Maintaining neutral posture keeps joints aligned and reduces muscle strain. Even light tasks can cause injury when posture is poor for extended periods.

- Head upright with ears aligned over shoulders
- Shoulders relaxed, not hunched
- Back supported with natural curves
- Elbows close to the body
- Wrists straight and neutral

Repetition, Movement, and Breaks

Repeating the same motion or holding the same position for long periods increases strain on muscles and joints. Movement helps restore circulation and reduces fatigue.

- Take short micro-breaks every 30–60 minutes
- Change positions frequently when possible
- Alternate tasks to reduce repetition
- Stretch hands, shoulders, neck, and back

Risk Factors & Prevention

Risk Factor	Prevention
Poor posture	Maintain neutral posture and proper workstation support
Repetitive motion	Rotate tasks and take regular micro-breaks
Static positions	Change position and stretch frequently
Improper lifting	Lift with legs and keep loads close to the body

Early Warning Signs & Reporting

Discomfort is an early warning sign. Reporting symptoms early allows adjustments to be made before injuries become serious or permanent.

- Persistent soreness or stiffness
- Tingling or numbness
- Reduced grip strength
- Limited range of motion
- Pain that worsens during work tasks

Final Reminder: Ergonomics is preventative safety. Daily habits shape long-term health. Addressing small discomforts early protects your body for the future.